Chapter 1 — Periodic Table and Periodic Properties

Exercise Solutions · Full textbook exercise · PECTAA Chemistry 11 · Book pp. 17–19

Q.1 — Multiple Choice Questions (with correct option and reason)

I. Which scientist first time observed the periodicity in the elements?

(a) J. Newlands
(b) L. Meyer
(c) J.W. Döbereiner
(d) D. I. Mendeleev

Answer: (a) J. Newlands — The textbook states that Newlands (1864) first observed periodicity when elements were arranged by increasing atomic mass; every eighth element resembled the first.

II. Recognize the element if it has 3 electron shells, belongs to "s" block and has 2 electrons in its outermost shell.

(a) Calcium
(b) Sodium
(c) Magnesium
(d) Potassium

Answer: (c) Magnesium — Three shells → Period 3; s-block with 2 valence electrons → Group 2. The element is Mg ($1s^2\ 2s^2\ 2p^6\ 3s^2$).

III. Which one do you think is correct about metallic character?

(a) It decreases from top to bottom in a group.
(b) It increases from top to bottom in a group.
(c) It remains constant from left to right in a period.
(d) It increases from left to right in a period.

Answer: (b) It increases from top to bottom in a group. — Down a group atomic size and shielding increase, so electrons are lost more easily and metallic character increases.

IV. Which property increases as you go down a group in the periodic table?

(a) Atomic radius
(b) Electron affinity
(c) Electronegativity
(d) Ionization energy

Answer: (a) Atomic radius — A new principal shell is added down a group, so atomic radius increases; the other three generally decrease.

V. Which set of the following conditions results in higher ionization energy?

(a) Smaller atom and greater nuclear charge.
(b) Smaller atom and smaller nuclear charge.
(c) Larger atom and greater nuclear charge.
(d) Larger atom and smaller nuclear charge.

Answer: (a) Smaller atom and greater nuclear charge. — A smaller radius and higher effective nuclear charge hold the outer electron more strongly, so more energy is needed to remove it.

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VI. Which of the following atoms show more than one (variable) oxidation states?

(a) Sodium
(b) Magnesium
(c) Aluminum
(d) Phosphorous

Answer: (d) Phosphorous — Na, Mg and Al show fixed +1, +2 and +3 states. Phosphorus shows +3 and +5 (e.g. PCl_3 and PCl_5).

VII. Which is the correct general trend in the variation of electron affinity in a group?

(a) It becomes less negative from top to bottom.
(b) It becomes more negative from top to bottom.
(c) It remains the same.
(d) It has no definite trend and changes irregularly.

Answer: (a) It becomes less negative from top to bottom. — Down a group the added electron is farther from the nucleus and more shielded, so electron affinity generally becomes less negative (halogens show the F/Cl exception).

VIII. What is the oxidation state of sulfur in the sulfate ion (SO_4^{2-})?

(a) +4
(b) +2
(c) +6
(d) 0

Answer: (c) +6 — $x + 4(-2) = -2 \rightarrow x = +6$.

IX. Which is the correct trend in variation of electronegativity along a period of the periodic table?

(a) It decreases from left to right across a period.
(b) It increases from left to right across a period.
(c) It remains constant.
(d) It has no definite trend.

Answer: (b) It increases from left to right across a period. — Across a period nuclear charge increases and atomic radius decreases, so the atom attracts the shared pair more strongly.

X. The atomic radius generally ..... across a period in the periodic table.

(a) Increases
(b) Decreases
(c) Remains constant
(d) First increases then decreases

Answer: (b) Decreases — Electrons are added to the same shell while protons increase, so effective nuclear charge pulls the cloud closer.

XI. Which one of the following elements has the highest ionization energy?

(a) Sodium (Na)
(b) Magnesium (Mg)
(c) Aluminium (Al)
(d) Argon (Ar)

Answer: (d) Argon (Ar) — Ar has a stable filled octet and the smallest radius among these Period-3 elements, so it has the highest first ionization energy.

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Q.2 — Short Answer Questions

a. What is 1st ionization energy? Give an example.

First ionization energy is the minimum energy required to remove one electron from one mole of gaseous atoms to form one mole of gaseous +1 ions. Example: $Na(g) \rightarrow Na^{+}(g) + e^{-}$; $I_1 = 496$ kJ mol^{-1}.

b. Explain why sulfur has a lower first ionization energy than phosphorus.

P has the stable half-filled $3p^3$ configuration. In S ($3p^4$) one 3p orbital contains a paired electron; electron–electron repulsion makes it easier to remove that electron. Hence $I_1(S) < I_1(P)$.

c. Why the elements in Group 13 to 17 are called p-block elements?

In these elements the last (differentiating) electron enters a p subshell of the outermost shell. The block name is taken from the subshell that receives the last electron, so Groups 13–17 (and 18) are p-block.

d. What are the factors that affect electronegativity?

(i) Atomic size: larger atoms have lower electronegativity because the bonding pair is farther from the nucleus. (ii) Effective nuclear charge: higher nuclear charge increases electronegativity. (iii) Shielding/screening: more inner shells reduce the attraction for the shared pair. (iv) Oxidation state and hybridisation: higher oxidation state and greater s-character increase electronegativity.

e. What factors are responsible for the increasing reactivity of alkali metals as we move down the group?

Down Group 1 atomic size and shielding increase, so the outer ns^1 electron is less strongly held and ionization energy decreases. The electron is lost more easily, making the metal more reactive (Cs > Rb > K > Na > Li).

f. Why some of the elements show variable oxidation numbers while others do not?

Elements with electrons of similar energy in ns, np and (n−1)d subshells can use different numbers of electrons in bonding, giving variable oxidation states (e.g. P: +3, +5; Fe: +2, +3). Elements with a single stable valence configuration (Na, Mg, Al) lose a fixed number of electrons and show only one common oxidation state.

g. Identify the element which is in period 5 and group 15.

Period 5, Group 15 is antimony, Sb (Z = 51). Its outer configuration is $5s^2\ 5p^3$.

h. Why oxides of sodium and magnesium are more ionic than the oxides of nitrogen and phosphorous?

Na and Mg are metals with low electronegativity, so the electronegativity difference with oxygen is large and electrons are transferred, forming ionic oxides (Na_2O, MgO). N and P are non-metals with electronegativity closer to oxygen, so they share electrons and form covalent oxides (N_2O_5, P_4O_{10}).

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i. Give reason for the different chemical reactivities of Na and Mg toward oxygen and chlorine.

Na has one valence electron and a lower ionization energy than Mg, which must lose two electrons. Therefore Na reacts more vigorously with O_2 and Cl_2. Both form ionic products: $4Na + O_2 \rightarrow 2Na_2O$ (or $2Na + O_2 \rightarrow Na_2O_2$ in excess air), $2Na + Cl_2 \rightarrow 2NaCl$; $2Mg + O_2 \rightarrow 2MgO$, $Mg + Cl_2 \rightarrow MgCl_2$.

j. Why the ionization energy of lithium is much lower than that of helium despite the fact that the nuclear charge of lithium is +3 and that of helium is +2?

He's electron is in the compact 1s shell, very close to the nucleus and part of a stable filled shell. Li's outer electron is in the larger 2s shell, shielded by the $1s^2$ core, so it is much easier to remove despite the higher nuclear charge.

k. The ionization energy of Be (atomic no. 4) is higher than that of B (atomic no. 5), despite the fact that the nuclear charge of Be is +4 and that of B is +5.

Be has a filled, stable $2s^2$ subshell. In B the outer electron is in a higher-energy 2p orbital, which is slightly shielded by the 2s electrons and is easier to remove. Hence $I_1(Be) > I_1(B)$.

l. What is common in Na^+, Mg^{2+}, Al^{3+}, Ne^0 and F^-? Arrange them in increasing order of sizes.

All are isoelectronic: each has 10 electrons ($1s^2\ 2s^2\ 2p^6$). For isoelectronic species, radius decreases as nuclear charge increases. Increasing size: $Al^{3+} < Mg^{2+} < Na < Ne < F^-$.

m. Consider the chlorides of sodium, magnesium, and phosphorus(V): NaCl, $MgCl_2$, and PCl_5. (i) Classify each as acidic, basic, or neutral. (ii) Explain referring to behaviour in water.

(i) NaCl: neutral; $MgCl_2$: essentially neutral (very slightly acidic, pH ≈ 6.5); PCl_5: acidic.

(ii) NaCl dissolves without hydrolysis: $NaCl(s) \rightarrow Na^+(aq) + Cl^-(aq)$, pH ≈ 7. $MgCl_2$ dissolves to hydrated Mg^{2+} and Cl^-; the textbook lists pH ≈ 6.5 and treats Group 1/2 chlorides as neutral. PCl_5 hydrolyses vigorously: $PCl_5 + 4H_2O \rightarrow H_3PO_4 + 5HCl$, giving an acidic solution.

Textbook check (p. 16): $MgCl_2$ is listed under "neutral chlorides" but its printed pH is 6.5, i.e. very slightly acidic due to weak hydrolysis of Mg^{2+}.

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Descriptive / Long Questions

Q.3 Write equations for the reactions of Na and Mg with oxygen, chlorine, and water. Compare the reactivity of both elements with these in terms of metallic character.

Sodium is more metallic (more electropositive) than magnesium because it loses its single 3s electron more easily. Consequently Na reacts more vigorously with water, oxygen and chlorine.

With water: $2Na(s) + 2H_2O(l) \rightarrow 2NaOH(aq) + H_2(g)$ (vigorous, cold water). Mg reacts slowly with cold water: $Mg(s) + 2H_2O(l) \rightarrow Mg(OH)_2(aq) + H_2(g)$; with steam it reacts faster: $Mg(s) + H_2O(g) \rightarrow MgO(s) + H_2(g)$.

With oxygen: sodium burns with a golden-yellow flame forming a mixture of oxide/peroxide; in open air peroxide is favoured: $2Na + O_2 \rightarrow Na_2O_2$; in limited O_2: $4Na + O_2 \rightarrow 2Na_2O$. Magnesium burns with an intense white flame: $2Mg + O_2 \rightarrow 2MgO$.

With chlorine: $2Na + Cl_2 \rightarrow 2NaCl$ (white solid, exothermic); $Mg + Cl_2 \rightarrow MgCl_2$ (white solid).

The greater ease of electron loss by Na (lower ionization energy, one valence electron) explains its higher reactivity compared with Mg, which must lose two electrons.

Textbook check (p. 14): the printed peroxide equation $2Na + O_2 \rightarrow 2Na_2O_2$ is unbalanced (correct: $2Na + O_2 \rightarrow Na_2O_2$); MgO is printed as MgO(g) but should be MgO(s); the steam equation is printed with $2H_2O/2H_2$ but balances as $Mg + H_2O \rightarrow MgO + H_2$.

Q.4 Explain with the help of equations, acidic and basic behavior of oxides and chlorides.

Basic oxides (metal oxides) react with water to give alkalis and with acids to give salt + water: $Na_2O + H_2O \rightarrow 2NaOH$; $CaO + H_2O \rightarrow Ca(OH)_2$; $MgO + 2HCl \rightarrow MgCl_2 + H_2O$.

Acidic oxides (non-metal oxides) react with water to give acids and with bases to give salt + water: $SO_2 + H_2O \rightarrow H_2SO_3$; $SO_3 + H_2O \rightarrow H_2SO_4$; $P_4O_{10} + 6H_2O \rightarrow 4H_3PO_4$; $SO_2 + 2NaOH \rightarrow Na_2SO_3 + H_2O$; $SiO_2 + 2NaOH \rightarrow Na_2SiO_3 + H_2O$.

Amphoteric oxides react with both acids and bases: $Al_2O_3 + 6HCl \rightarrow 2AlCl_3 + 3H_2O$; $Al_2O_3 + 2NaOH \rightarrow 2NaAlO_2 + H_2O$.

Neutral chlorides: NaCl and $MgCl_2$ dissolve without appreciable hydrolysis: $NaCl \rightarrow Na^+ + Cl^-$ (pH ≈ 7).

Acidic chlorides: covalent chlorides hydrolyse to give HCl: $AlCl_3 + 3H_2O \rightarrow Al(OH)_3 + 3HCl$; $SiCl_4 + 2H_2O \rightarrow SiO_2 + 4HCl$; $PCl_5 + 4H_2O \rightarrow H_3PO_4 + 5HCl$. The solution becomes acidic (pH < 7).

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Q.5 Describe the factors affecting and periodic trends of electron affinity.

Electron affinity is the energy change when one mole of gaseous atoms gains electrons to form one mole of gaseous negative ions: $X(g) + e^- \rightarrow X^-(g)$. A more negative value means greater tendency to accept an electron.

Factors: (i) Atomic size — smaller atoms attract the added electron more strongly, giving more negative electron affinity. (ii) Effective nuclear charge — higher nuclear charge increases attraction. (iii) Shielding — more inner shells reduce attraction. (iv) Electronic configuration — stable half-filled (N, P) or fully filled (noble gases) subshells resist addition, giving low or positive values.

Across a period: electron affinity generally becomes more negative from left to right because nuclear charge increases and size decreases. Exceptions: Group 2 (filled s) and Group 15 (half-filled p) have low values; noble gases have positive values.

Down a group: electron affinity generally becomes less negative because size and shielding increase. Exception: Cl (-349 kJ mol^{-1}) is more negative than F (-328 kJ mol^{-1}) because the added electron in very small F experiences strong repulsion in the compact 2p shell.

Trend	General behaviour	Reason
Across a period	More negative (left → right)	Higher Zeff, smaller size
Down a group	Less negative	Larger size, more shielding
F vs Cl	Cl more negative than F	Strong e^-–e^- repulsion in compact F 2p
N, P / noble gases	Low or positive	Stable half-filled / filled subshells

Q.6 Define ionization energy. Discuss the factors affecting and periodic trends of ionization energy.

Ionization energy is the minimum energy required to remove the most loosely bound electron from one mole of gaseous atoms to form one mole of gaseous +1 ions: $M(g) \rightarrow M^+(g) + e^-$. Successive ionization energies (I_1, I_2, I_3…) refer to removal of the 1st, 2nd, 3rd… electrons.

Factors: (i) Atomic size — larger atoms have lower I.E. (ii) Nuclear charge — higher charge increases I.E. (iii) Shielding — more inner shells lower I.E. (iv) Penetration — s electrons are held more tightly than p, p more than d. (v) Stability of configuration — filled/half-filled subshells raise I.E.

Across a period: I.E. generally increases because size decreases and nuclear charge increases. Exceptions: Be > B (2s vs 2p) and N > O (half-filled $2p^3$ vs paired $2p^4$).

Down a group: I.E. decreases because a new shell is added and shielding increases, so the outer electron is easier to remove.

Trend	General behaviour	Important exceptions
Across a period	Increases	Be > B; N > O
Down a group	Decreases	—
Successive I.E.	$I_1 < I_2 < I_3 \ldots$	Large jump after noble-gas core is reached

Quick Check Solutions (in-chapter practice boxes)

Quick Check 1.1 (book p. 5)

a. Why are the elements in Groups 1 and 2 known as s-block elements?

Their last (differentiating) electron enters an s subshell of the outermost shell (ns^1 for Group 1, ns^2 for Group 2), so they are placed in the s-block.

b. Name the elements in the chalcogen family. Give their two characteristics.

Chalcogens are Group 16: O, S, Se, Te, Po. Characteristics: (i) six valence electrons ($ns^2\ np^4$), so they tend to gain/share two electrons; (ii) they form acidic oxides and hydrides of type H_2E.

Quick Check 1.2 (book p. 6)

a(i–ii). X belongs to group 14 and period 2: write configuration, block and identity.

Period 2 + Group 14 → carbon (C, Z = 6). Configuration: $1s^2\ 2s^2\ 2p^2$. It is a p-block element.

b. Predict the electronic configuration of an element in period 4 and group 17.

Period 4, Group 17 is bromine (Br, Z = 35): $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}\ 4s^2\ 4p^5$, or [Ar] $3d^{10}\ 4s^2\ 4p^5$.

Quick Check 1.3 (book p. 7)

a. Which factors affect atomic and ionic radii?

Number of shells, effective nuclear charge, shielding by inner electrons, and (for ions) the charge gained or lost.

b(i). Relative sizes: atomic radii of lithium and fluorine.

Li > F. Across Period 2, nuclear charge increases while electrons enter the same shell, so F is smaller.

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b(ii). A lithium atom and its ion, Li^+.

Li > Li^+. Li^+ loses the 2s shell, leaving the smaller $1s^2$ core.

b(iii). An oxygen atom and its ion, O^{2-}.

O^{2-} > O. Added electrons increase electron–electron repulsion, expanding the cloud.

b(iv). A nitride ion, N^{3-}, and a fluoride ion, F^-.

N^{3-} > F^-. They are isoelectronic (10 e^-); N has fewer protons (7 vs 9), so it holds the electrons less tightly.

Quick Check 1.4 (book p. 10)

a(i). 1st ionization energy of boron is lesser than beryllium.

Be removes a 2s electron from a filled, stable $2s^2$ subshell; B removes a higher-energy 2p electron, which is easier to remove.

a(ii). 1st ionization energy of aluminum is lower than magnesium.

Mg removes a 3s electron from a filled $3s^2$ subshell; Al removes a higher-energy 3p electron, so I_1(Al) < I_1(Mg).

b. Trend in ionization energy down group 3 (13). Give reason.

It decreases down the group because atomic size and shielding increase, so the outer electron is less strongly held.

Quick Check 1.5 (book p. 11)

a. 1st electron affinity of oxygen is −141 kJ/mol but 2nd is +844 kJ/mol.

The first electron is attracted to the neutral O atom, releasing energy. The second electron is repelled by the already negative O^- ion, so energy must be supplied.

b. Which of nitrogen and phosphorus has the higher electron affinity? Justify.

Phosphorus has the higher (more negative) electron affinity. Nitrogen's compact, half-filled $2p^3$ subshell strongly resists adding an electron; P's larger 3p subshell accepts it more readily.

c. F has lower electron affinity than Cl although its size is smaller.

The added electron in very small F enters a compact 2p shell and experiences strong electron–electron repulsion, reducing the energy released compared with Cl.

d. Why noble gases have positive 1st electron affinities.

They have stable filled octets; an added electron must enter a higher-energy shell, so energy is absorbed rather than released.

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Quick Check 1.6 (book p. 13)

a. Illustrate how metallic character varies in group 14.

It increases down the group: C is a non-metal, Si and Ge are metalloids, Sn and Pb are metals, because ionization energy decreases and electrons are lost more easily.

b. Identify semi-metals in groups 14, 15 and 16. Why are they semi-metals?

Group 14: Si, Ge; Group 15: As, Sb; Group 16: Te (Po is often classed as a metal). They show intermediate properties between metals and non-metals because their ionization energies/electronegativities are intermediate.

Quick Check 1.7 (book p. 14)

a. Nature of oxides and hydroxides of Na and Mg.

Na_2O/NaOH and MgO/$Mg(OH)_2$ are basic. $Na_2O + H_2O \rightarrow 2NaOH$; $MgO + H_2O \rightarrow Mg(OH)_2$ (slow). They react with acids to form salts and water.

b. Predict the reactivity of Ca with water and oxygen.

Ca is more reactive than Mg (lower I.E.). It reacts with cold water to give $Ca(OH)_2 + H_2$, and burns in oxygen to form CaO (and some Ca_3N_2 in air).

Quick Check 1.8 (book p. 16)

a. ZnO reacts with HCl and NaOH. Give equations and predict the type of oxide.

$ZnO + 2HCl \rightarrow ZnCl_2 + H_2O$; $ZnO + 2NaOH \rightarrow Na_2ZnO_2 + H_2O$. Since it reacts with both acid and base, ZnO is amphoteric.

b. Why $AlCl_3$ is an acidic halide, but NaCl not.

$AlCl_3$ is covalent and hydrolyses in water to give HCl (acidic): $AlCl_3 + 3H_2O \rightarrow Al(OH)_3 + 3HCl$. NaCl is ionic and simply dissolves without hydrolysis, giving a neutral solution.

c. Predict whether PCl_5 and NCl_3 would be acidic or basic.

PCl_5 is acidic: $PCl_5 + 4H_2O \rightarrow H_3PO_4 + 5HCl$. NCl_3 hydrolyses to NH_3 and HOCl ($NCl_3 + 3H_2O \rightarrow NH_3 + 3HOCl$); the NH_3 makes the solution basic.

d. Would SO_2 and P_2O_5 react with HCl/H_2SO_4 or with NaOH?

They are acidic oxides, so they react with the base NaOH, not with acids. $SO_2 + 2NaOH \rightarrow Na_2SO_3 + H_2O$; $P_2O_5 + 6NaOH \rightarrow 2Na_3PO_4 + 3H_2O$.

Quick Check 1.9 (book p. 17)

a. Calculate the oxidation number of sulfur in SO_2 and SO_3.

SO_2: $x + 2(-2) = 0 \rightarrow x = +4$. SO_3: $x + 3(-2) = 0 \rightarrow x = +6$.

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b. Why some p-block elements show variable oxidation state.

They can use different numbers of valence electrons in bonding (and, for Period 3 onward, involve d orbitals), giving states such as +3 and +5 for P, or +4 and +6 for S.

Textbook checks flagged in this chapter

• p. 14: $2Na + O_2 \rightarrow 2Na_2O_2$ is unbalanced; the balanced peroxide equation is $2Na + O_2 \rightarrow Na_2O_2$. MgO(g) should be MgO(s). The steam equation should be $Mg + H_2O \rightarrow MgO + H_2$.

• p. 16: $MgCl_2$ is grouped with “neutral chlorides” although the printed pH is 6.5 (very slightly acidic).

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